



ECOINCENTIVES PROTOCOL

An EVM-Compatible Smart Contract Protocol for the Circular Economy

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The Waste Crisis in Peru

A Market Failure Driven by Negative Externalities (*Pigou, 1920*). The private cost of disposing of waste informally is lower than the social cost.

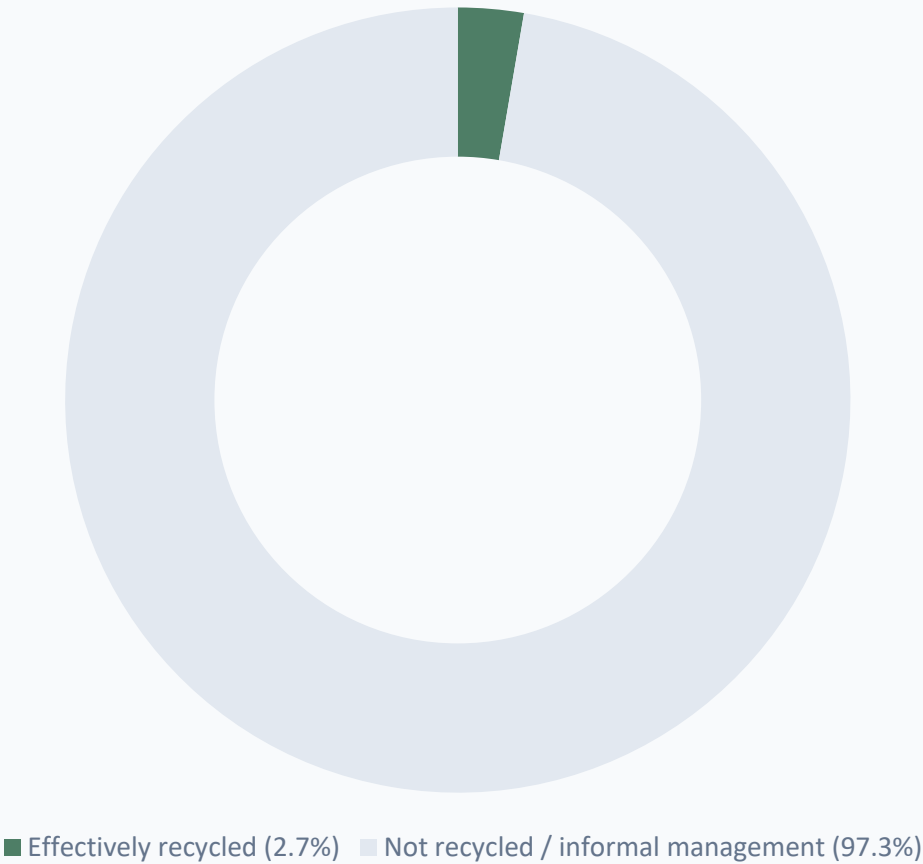
9.0M

Tonnes of municipal solid waste generated per year (MINAM, 2024)

2.7%

Formal recycling and recovery rate nationwide

Effective Recycling Rate — Peru (2024)



Why Do Traditional Initiatives Fail?



Transaction Costs

(Coase, 1960)

Sorting and transporting waste imposes a prohibitive opportunity cost in time and effort on households, with no compensation whatsoever.



Information Asymmetry

(Akerlof, 1970)

A two-way gap: citizens distrust the final destination of their waste, while municipalities lack predictable data on household-level recycling supply.



Enforcement Costs

Traditional centralized subsidies require massive manual audits, creating financial leakage and high administrative friction.

EcoIncentives: A Digital Pigouvian Subsidy

01



Source Segregation

The citizen sorts recyclable material at home, eliminating the centralized transaction cost.

02



Cryptographic Validation

The municipal operator weighs the material and logs the kilograms on the protocol's node.

03



On-Chain Minting

The smart contract automatically mints EcoPoints (ECO) directly to the user's wallet.

Blockchain Architecture & Immutability



Solidity 0.8.20

A robust, self-executing smart contract, natively deployed on LNet (LACNet Pro-Testnet).



O(1) Gas Optimization

Structured storage via direct mappings instead of loops over dynamic arrays, preventing block-space saturation.



Access Controls

Restrictive modifiers (`onlyOwner`) ensure that only authorized municipal operators can mint points after a real physical validation.



```
// EcoIncentives.sol - excerpt
function registerRecycling(
    address citizen, uint256 kg
) external onlyOwner {
    require(kg > 0, "Invalid input");
    uint256 pts = kg * RATE;
    balances[citizen] += pts;
    emit RecyclingVerified(citizen, kg, pts);
}
```

Adoption Without Technological Barriers



Web3 Authentication

Direct integration with the MetaMask API. Secure one-click access, with no centralized passwords.



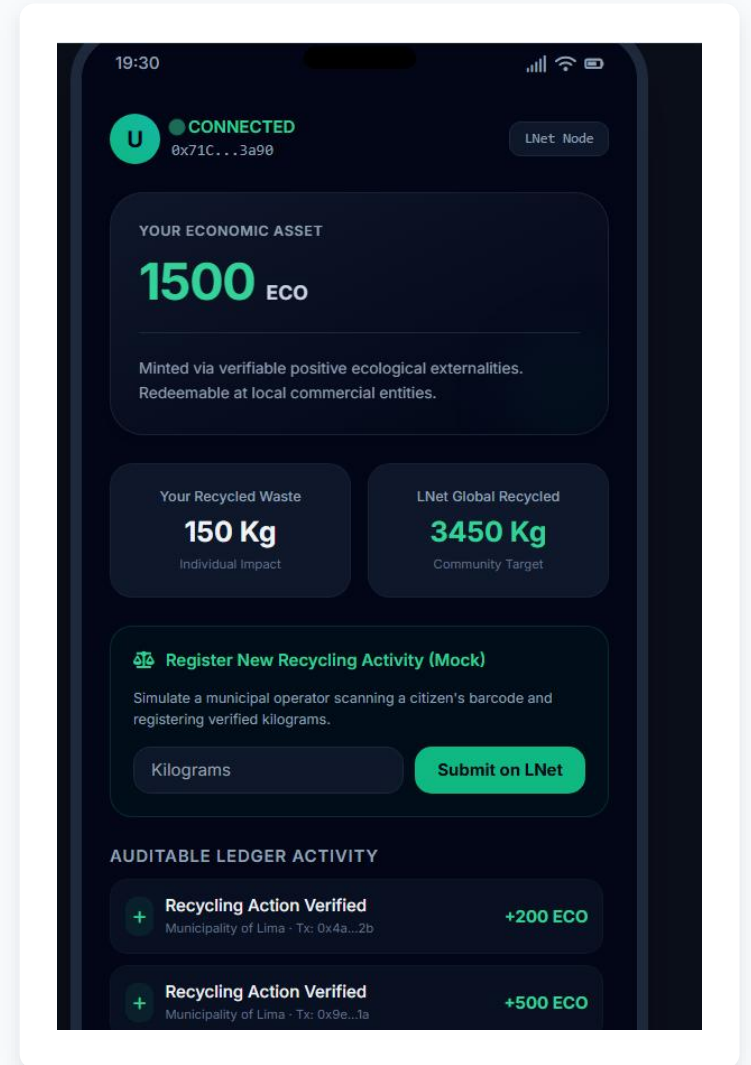
Eco Dashboard

Real-time interactive visualization of the ECO token balance and impact metrics (cumulative kg recycled).

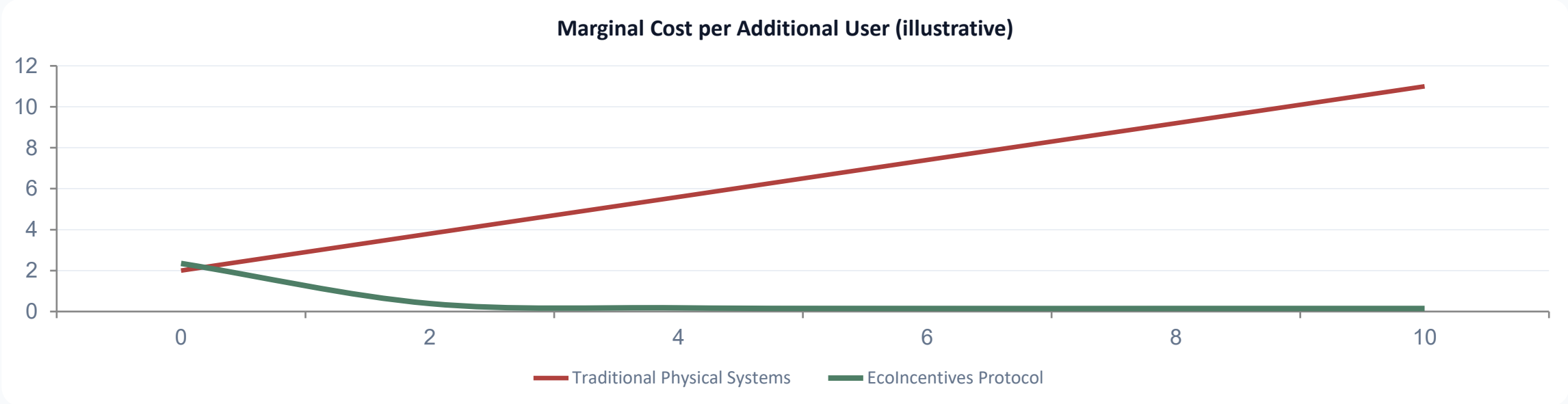


EcoMarketplace

An integrated interface to execute redeemReward, securely burning tokens to claim discounts at affiliated merchants.



Cost Structure & Signaling Theory



Marginal Cost Tending to Zero ($MC \rightarrow 0$)

Expanding the system to new districts or thousands of users requires no heavy physical infrastructure; the protocol scales instantly on LNet's existing distributed nodes.



Signaling Theory (Spence, 1973)

The public, immutable blockchain record acts as a credible, auditable green signal, letting businesses verify a customer's real ecological behavior without costly audits.

Projection & Deployment Plan

Q3 2026



Technical Rigor & Security

Comprehensive vulnerability audit of the smart contract, plus an additional 15% reduction in gas fee usage.

Q4 2026



Real Market Validation

Launch of a pilot program in a selected district of Lima, Peru. Target: 500 active households and 2 metric tons processed.

Q1 2027



Commercial Network Expansion

Integration of commercial APIs to redeem ECO tokens at supermarket chains.
Deployment across 2 additional municipalities.

Open-Source Software as a Regional Public Good

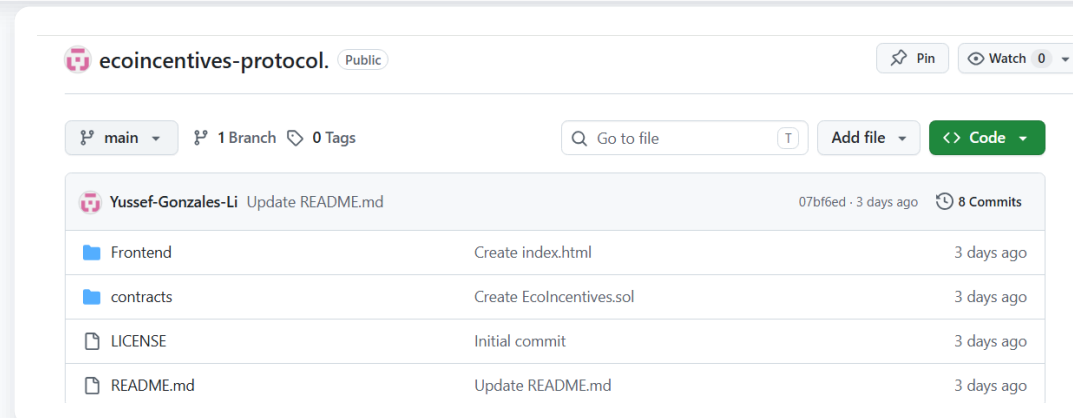
Built on the principles of non-rivalry and non-excludability to enable cumulative innovation across municipalities in Latin America.



Official GitHub Repository: `github.com/Yussef-Gonzales-Li/EcoIncentives`



Protocol Demo Video: `video.ecoincentives.protocol`



Thank You!